

Available online at www.sciencedirect.com**SciVerse ScienceDirect**journal homepage: www.elsevier.com/locate/ijrefrig

Falling film evaporation on enhanced tubes, part 1: Experimental results for pool boiling, onset-of-dryout and falling film evaporation

Marcel Christians, John Richard Thome*

Laboratory of Heat and Mass Transfer (LTCM), Ecole Polytechnique Fédérale de Lausanne, ME G1 465, Station 9, CH-1015 Lausanne, Switzerland

ARTICLE INFO

Article history:

Received 9 September 2010

Received in revised form

25 October 2011

Accepted 31 October 2011

Available online 11 November 2011

Keywords:

Evaporation

Falling film

Pool boiling

Two-phase flow

ABSTRACT

Horizontal falling film evaporators have the potential of displacing flooded evaporators as the standard in the refrigeration industry due to compelling advantages, such as smaller refrigerant charge and lower pressure drop. However, there is a need to improve the understanding of falling film evaporation mechanisms to provide accurate thermal design methods. In this work, falling film evaporation measurements on a single tube, a vertical row of ten horizontal tubes and a small tube bundle with three rows of 10 tubes each are presented. Two enhanced boiling tubes, the Wolverine Turbo-B5 and the Wieland Gewa-B5, were tested using R-134a and R-236fa. The tests were carried out at a saturation temperature of $T_{\text{sat}} = 5^\circ\text{C}$, film Reynolds numbers ranging from 0 to 3000, and heat fluxes between 15 and 90 kW m^{-2} in pool boiling and falling film configurations. A visualization study was performed under diabatic conditions to study the flow.

© 2011 Elsevier Ltd and IIR. All rights reserved.

Evaporation d'un film tombant sur des tubes améliorés, Partie 1 : résultats expérimentaux pour l'ébullition libre, le début de l'assèchement et l'évaporation d'un film tombant

Mots clés : Évaporation ; Film tombant ; Ébullition libre ; Écoulement diphasique

1. Introduction

In industrial applications, falling film evaporators show several advantages compared to flooded evaporators, particularly in terms of higher cycle efficiency, reduced costs and

a smaller environmental impact due to its reduced charge of refrigerant. The pressure drop is small as the liquid flows only due to gravity and this reduces the temperature lift of the compressor. This type of heat exchanger has been studied over the years in terms of effects such as the liquid feed flow rate,

* Corresponding author. Tel.: +41 21 693 5981; fax: +41 21 693 5960.

E-mail address: john.thome@epfl.ch (J.R. Thome).

0140-7007/\$ – see front matter © 2011 Elsevier Ltd and IIR. All rights reserved.

doi:10.1016/j.ijrefrig.2011.10.020

Nomenclature		Greek	
Symbol		Γ	ref. overfeed per unit length ($\text{kg m}^{-1} \text{s}^{-1}$)
c_p	isobaric specific heat ($\text{J kg}^{-1} \text{K}^{-1}$)	Subscripts	
C	Wilson plot coefficient	ff	falling film
D	diameter (m)	gni	Gnielinski
h	heat transfer coefficient ($\text{W m}^{-2} \text{K}^{-1}$)	i	internal
K_{ff}	Falling film multiplier ($h_{ff} h_{pb}^{-1}$)	o	outer
$\dot{m}$	water mass flow rate (kg s^{-1})	pb	pool boiling
n	number of datapoints	sat	saturation
q	heat flux (W m^{-2})	$water$	water
R_w	wall thermal resistance (K W^{-1})	Superscripts	
T	temperature ($^{\circ}\text{C}$)	n	pool boiling exponent
U	overall heat transfer resistance (K W^{-1})		

the liquid distribution, the flow pattern, liquid subcooling, tube surface phenomena, tube spacing and heat flux, among others. A large majority of the studies have been performed using water or ammonia, due to the large potential of falling film evaporators in Ocean Thermal Energy Conversion and actual use in desalination. Falling film evaporation can also be segregated into conditions with or without nucleate boiling in the film. The present study considers only the former.

Recent studies involving pool boiling at conditions similar to those being investigated in this study are presented in Table 1. The majority of these studies have been discussed by Roques and Thome (2007a). There are three main additions to the studies discussed previously, namely those of Habert (2009), Agostini (2008) and Gorgy (2008).

Gorgy (2008) tested four different combinations: R-134a on a smooth tube, R-123 on a smooth tube, R-134a on a Turbo-Bii HP tube, and R-123 on a Turbo Bii-LP tube. A modified Wilson plot method was utilized to measure the heat transfer coefficients. The one meter long test tube's outer diameter was 19.05 mm. Tests were conducted with a single tube test section, in which the test tube was water-heated. All tests were conducted at a saturation temperature of 4.44 $^{\circ}\text{C}$. Results showed that the heat transfer coefficient increased with heat flux for all cases.

Habert (2009) tested the pool boiling performance of the Turbo-EDE2, Gewa-C LW, Gewa-B4 and a plain tube using refrigerants R-134a and R-236fa at saturation temperatures of 5, 10 and 20 $^{\circ}\text{C}$. The tubes tested were 19.05 mm in diameter, and 0.554 m long. An indirect method was utilized to infer the pool boiling heat transfer coefficient (modified Wilson Plot method). Using the plain tube, the study generated results similar to those expected from the literature; the heat transfer measurements, the values of the correlations and the trends are similar to the results of citetstephan1992, Stephan and Abdelsalam (1980) and Cooper (1984). The Gewa-C LW, a tube designed for condensation, behaved in a similar fashion. The two enhanced evaporation tubes, on the other hand, showed decreasing heat transfer with increasing heat flux. The trend of these results are similar to those found by Roques and Thome (2007a).

Boiling in a thin film differs from its pool boiling counterpart. Falling film evaporation provides higher heat transfer coefficients than pool boiling in the low heat flux convective region. Cerza and Sernas (1985) hypothesized that the enhancement of the nucleate boiling heat transfer in liquid films is due to the fact that the bubbles lie embedded in the superheated liquid film, compared to nucleate boiling where bubble growth is generally confined

Table 1 – Recent pool boiling studies at conditions close to those in this study. A:Plain, B:Turbo-Bii HP & LP, C:Gewa-B, D:High-Flux, E:Turbo-B, F:Turbo-EDE2, G:Gewa-B4, H:Gewa-C LW, I:Turbo-CSL.

Authors	Tube	d [mm]	Fluid	$\dot{q}$ [kW m^{-2}]	T [K]
Tatara and Payvar (2000)	B	19.05	R-134a	8–40	278
Hahne and Barthau (2006)	A	15	R-134a	1–100	282–356
Danger et al. (2001)	A	25.4	R-134a	1–100	263–282
Robinson and Thome (2003)	A, B	19.05	R-410a, R-507a	15–40	278
Hsieh and Hsu (1994)	A	27	R-134a	8–60	278
Webb and Pais (1992)	A, E	19.1	R-134a	8–60	278
McNeil et al. (2002)	A, D	19	Pentane	5–50	309
Memory et al. (1995)	A, D, E	15.9	R-114	1–100	275
Kramer et al. (1997)	A, C	22.2	H_2O , Ethanol	15–65	351–373
Roques and Thome (2007a)	A, B, C, D	19.05	R-134a	25–65	278
Habert (2009)	A, F, G, H	19.05	R-134a, R-236fa	20–80	278–293
Gorgy (2008)	A, B	19.05	R-134a, R-123	4–120	278
Agostini (2008)	A	19.05	R-134a, R-236fa	15–70	283–293
Ribatski and Thome (2006)	B, C, D, I	19.05	R-134a	15–80	278–293

to the thickness of the superheated thermal boundary layer next to the tube wall.

The enhanced boiling process is different from normal nucleate boiling in cavities. Evaporation occurs both on the external surface, and inside its re-entrant channels. Thus, there are four possible ways for heat to be dissipated (Bergles, 1988; Thome, 1990).

1. As latent heat generating vapor within the re-entrant channels.
2. As latent heat into bubbles growing on the outside surface of the enhancement.
3. As sensible heat to the liquid, pumped through the re-entrant channels and back out.
4. As sensible heat to the liquid in the external thermal boundary layer.

Structured surfaces promote nucleate boiling in the film at modest temperature differences, enhance convection within the film and provide an increase in heat transfer area. The parameters that influence the enhancement are mainly the shape, geometry and surface area of the cavities. The nucleation site density also plays an important role. For porous coatings, the principal parameters are the particle size, the coating thickness and the porosity. The nucleation site density has been correlated as proportional to the heat transfer coefficient. Chien and Webb (1998a, b) investigated the effect of pore diameter, pore pitch and tunnel shape using R-11 and R-123. They found that there exists an optimal pore diameter and pore pitch for a specific heat flux range. However, the determination of the above parameters is purely empirical, and difficult to ascertain (Poniewski and Thome, 2008).

In comparison to pool boiling, experimental data for boiling in thin films are relatively scarce, as shown in the reviews of Thome (2002) and Ribatski and Jacobi (2005). Chyu and Bergles (1985) investigated falling film heat transfer mechanisms and defined three flow regimes. For the case of saturated falling films, convective heat transfer leads to evaporation at the liquid–vapor interface. With an increase of heat flux, nucleate boiling occurs; vapor bubbles grow and are carried along by the film flow such that both thin falling film evaporation and nucleate boiling play a role in the heat transfer process, depending mainly on the heat flux and liquid mass flow. Moeykens and Pate (1994) and Zeng et al. (1996) among others published mean heat transfer coefficient data for each tube in their row/bundle tests.

Chien and Webb (1998a, b) tested enhanced surfaces similar to the Turbo-B using R-11 and R-123. It was observed that, at low heat flux, the tubes having smaller total open areas (that is, the sum of the cavity areas) gave higher heat transfer coefficients. At higher heat fluxes, tubes having larger total open areas yielded higher heat transfer performance. They reported a cross-over characteristic between the two areas. If the total open area was too large at low heat flux, the re-entrant tunnels became flooded by liquid and the heat transfer coefficient decreased. If the total open area was too small at high heat fluxes, the tunnels dried out due to inadequate liquid supply. Chien and Webb (1998c) also performed a visualization study that supported these trends.

Moeykens et al. (1995) observed that enhanced boiling surfaces gave higher performance than finned tubes but lower performance than enhanced condensing surfaces used for evaporation. They noted an increase of heat transfer coefficient with heat flux up to a specific heat flux, after which, with any further increase in heat flux, the heat transfer coefficient decreased. This was probably due to partial dryout.

Roques and Thome (2007a, b) tested three different enhanced surfaces: Gewa-B, Turbo-Bii and High-Flux. Similar trends for each surface were found, as well as a strong dependence of heat transfer on heat flux. The High-Flux tube achieved up to three times better performance than the other tubes tested. A falling film multiplier, K_{ff} , was defined as the ratio between falling film evaporation and pool boiling heat transfer coefficients on the same tube at the same heat flux, and gave values between 1 and 2, depending on the enhanced surface and the experimental conditions, meaning that the film had a tendency to enhance the heat transfer performance. In general, at low heat fluxes, falling film achieves better performance than pool boiling. This might be due to enhanced convective effects in the falling film. At high heat fluxes, nucleate boiling was the dominant heat transfer mode and thus the convective effect tended to disappear and the performances became comparable to those in pool boiling. According to the authors, the trend for falling film evaporation is made up of two distinct regions: a plateau corresponding to an all-wet nucleate boiling-dominated regime where the heat transfer coefficient varies little with the flow rate, a point of onset-of-dryout, and a partially wet regime with nucleate boiling in the remaining film with a thus rapidly decreasing heat transfer coefficient tending towards the vapor phase natural convection coefficient at complete dryout. Hence, the onset-of-dryout is an important part of the heat transfer process and its modeling.

Habert and Thome (2010) tested three enhanced surfaces, the Gewa-B4, Turbo-EDE2 and the Gewa-C LW, a condensation tube. The tests were performed using R-134a and R-236fa. Tests in single-array mode, similar to those of Roques (2004), were performed. Using the Gewa-C LW, K_{ff} values of 2.5–3 were found near the onset-of-dryout Reynolds number using R-134a and R-236fa respectively, at a heat flux of 20 kW m^{-2} . However, most of the data lie close to 1, indicating similar performance under falling film and pool boiling conditions. The performance of the tube did not vary much with choice of refrigerant. Both the Gewa-B4 and the Turbo-EDE2 show similar trends to the previous generation's (that is, tubes tested by Roques); however, due to the improved performance, the uncertainty in the measurements also increased. Heat transfer coefficients close to those measured by Roques for the High-Flux tube were measured. The falling film multiplier varied between 1 and 2, with falling film performing best at lower heat fluxes, as shown in previous studies.

The objectives of the present study are to (i) characterize the water-side enhancement factor of different enhanced tubes using the modified Wilson plot method, (ii) measure local pool boiling coefficients at 5, 10 and 15°C for heat fluxes ranging from 20 to 80 kW m^{-2} , (iii) measure local heat transfer coefficients at 5°C on a one-row horizontal tube array and on a small three-row bundle for heat fluxes of 20, 40 and

60 kW m⁻², (iv) extend the existing database with new surfaces and fluids and (v) to develop new predictive methods for pool boiling, onset-of-dryout, the local heat transfer coefficient and the falling film multiplier K_{ff} . The results of the experimental campaign are presented in part 1 while the new prediction methods are presented in part 2 (Christians and Thome, 2010).

2. Experimental facility

The experimental setup is comprised of three circuits, namely, the refrigerant, water-glycol, and water circuits. The refrigerant circuit is shown schematically in Fig. 1. It comprises an electrically heated evaporator (1) to maintain the desired saturation condition, a condenser (5) to balance the heat load on the system, and the test section itself (4).

In the refrigerant circuit, starting from the flooded evaporator (1), the refrigerant flows through the filter (not shown) and the subcooler (6) to the oil-free self-lubricating gear pump (7). Parallel to the pump, bypass piping is installed so that, together with a frequency controller on the pump, the desired liquid flow rate can be accurately set. A Coriolis mass flow meter (8) follows, after which an electric heater (9) is installed to bring the liquid close to saturation conditions at the test section inlet (less than 0.5 K from saturation). At this point, the liquid enters the test section and is distributed uniformly on the top row of the heated tubes. Special care has been taken in the distributor design in order to achieve uniform liquid distribution on the top tube. Once the liquid leaves the distributor, it falls onto the top of the heated tube array on which the liquid in the test section is partially evaporated; the residual liquid leaves the test section by gravity. From the exit of the test section, the liquid flows back to the flooded

evaporator by the effect of gravity. The vapor generated in the test section is sucked out at the bottom of the test section. After exiting the test section from the bottom, it flows to the condenser, and the liquid drops by gravity back to the flooded evaporator. The amount of vapor flow can be controlled by increasing the heat input in the flooded evaporator, which in turn generates more vapor. Consequently, to maintain a constant system pressure, the cooling load on the auxiliary condenser is increased to balance the thermal heat load. In these tests, it was attempted to maintain the vapor velocity as low as possible, such that vapor shear effects were minimized.

The water circuit (not shown) is responsible for the heating effect in the test section. The water is driven through the test tubes by a centrifugal pump. An electronic speed-controller, together with a bypass line, ensures good precision in any water mass flow adjustment. The water flows through two liquid–liquid heat exchangers; the first is cooled with industrial water sourced from Lake Geneva at a constant temperature of 7 °C, while the second is heated with hot water from a closed-loop circuit heated by a heat pump. This water has its flow rate set by a computer-controlled valve. The water temperature at the test section inlet is thus automatically maintained constant. The total water mass flow rate is measured with a Coriolis flow meter (not shown). Before entering the test section, the test-line water flow is split into five subcircuits, each supplying to two tubes in the test section connected in series. Each subcircuit has two tube passes; i.e., water goes in the first tube in one direction (left to right) and comes back through the copper tube just above in the opposite direction. This setup keeps the film flow rate uniform along the axial length of the tube after each of the sets of two tubes to avoid maldistribution that occurs in one-pass test setups. A water-glycol mixture from a network installation is used as a cold source for the auxiliary condenser.

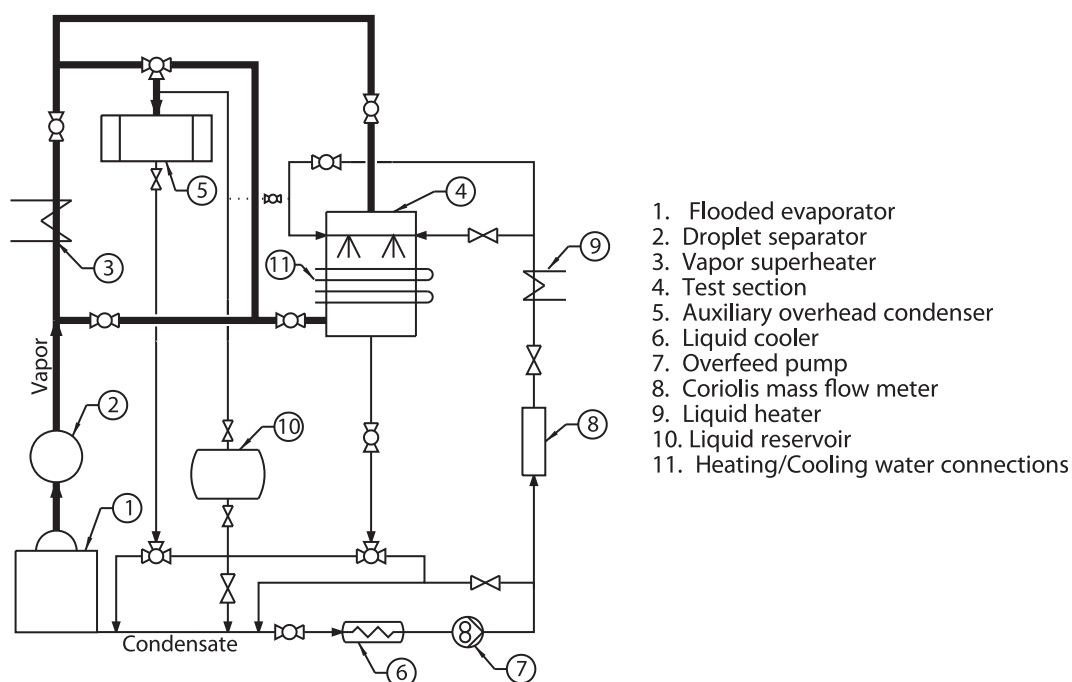


Fig. 1 – Schematic of the refrigerant circuit in the Falling Film Facility.

The test section is a rectangular stainless-steel vessel with six large windows situated at the front and rear in order to have full visual access into the experimental setup, to observe the flow on the tubes. The copper test tubes had a nominal outer diameter of 19.05 mm and a length of 554 mm. In total, ten instrumented tubes (i.e., five sub-circuits) were installed at an industry-standard pitch of 22.3 mm and arranged in one of two possible configurations. For single-array tests, these ten tubes were installed horizontally in a vertical array. For bundle tests, two additional vertical arrays of ten horizontal tubes were installed on either side of the central array using equal vertical and horizontal pitches (See Figure 3.7 of Christians (2010)). Falling film conditions were created on these two rows using two sets of two high-performance condensing tubes installed on both sides of the distributor, through which cold glycol is run. These four tubes provided the condensate overfeed for the side rows. The glycol mass flow rate was controlled to regulate the amount of condensate being generated. To further fine tune the amount of heat exchanged, a three-way valve is installed. The glycol flow rate through each side could be manually controlled such that there was no imbalance in the heat extraction between the two sides.

Hot water was pumped through the side-array tubes; however, unlike the center column, the water went through all the side tubes in ten passes (rather than the two passes in the center column for each pair of tubes). The water flow rate through each side-array could be controlled in a similar fashion to the glycol overfeed circuit, such that the water-heated side-array heat transfer remain balanced.

A stainless-steel tube with an external diameter of 8 mm was inserted inside each of the copper test tubes (in bundle conditions, only in the ten tubes in the central horizontal column). Pairs of thermocouples were located at three positions axially along the stainless steel tube, protruding out through holes to measure the temperature of the water in the annulus between the stainless steel tube and the copper tubes. At every location, one thermocouple is facing upward and one is facing downward. A copper wire with a rectangular cross section wound helically around the stainless steel tube promoted mixing, and further increased the water-side heat transfer coefficient.

Pressure transducers connected to the test section above and below the array of tubes were used to measure the vapor pressure in the test section. The vapor temperature in the test section was measured above and below the tube array using sheathed thermocouples. The temperatures of the liquid entering and leaving the test section, as well as the vapor leaving the test section, were measured.

3. Experimental methods and data reduction

The internally mounted thermocouples measuring the water temperature within the tube annulus along the axial length of the tubes provide the water temperature profile as a function of the distance x along the tubes. Assuming only heat flow in the radial direction, the local heat flux on the outside of the tube, q_o , may thus be expressed as

$$q_o = \frac{\dot{m}_{\text{water}} C_{p,\text{water}}}{\pi D_o} \left(\frac{dT_{\text{water}}}{dx} \right) \quad (1)$$

where D_o is the outside diameter. The value (dT_{water}/dx) is obtained by differentiating a second order polynomial fit of the water temperature profile. Nearly identical temperatures for the pairs of thermocouples located at each location were found, indicating good mixing of the water (the temperatures were within thermocouple uncertainty, i.e. ± 0.1 K), and thus increasing the accuracy of the data reduction method.

To determine the external local heat transfer coefficient, h_o , between the outside surface of the copper tubes and the refrigerant, a modified Wilson plot procedure using nucleate pool boiling on the outside of the tubes was first implemented to infer the internal heat transfer coefficient. The method utilized is described in detail in Christians (2010) and van Rooyen et al. (submitted for publication). The modified Wilson plot method used a relation for the external heat transfer coefficient given by $h_o = C_o q_o^n$, where n and C_o were left as variables. The internal heat transfer coefficient was given by the Gnielinski (1975) correlation, h_{gni} , multiplied by a constant C_i (also a variable) that takes into account the increase in heat transfer due to any internal enhancement, the reduced flow area and increased turbulence due to the inserted helical tape. The Wilson plot expression for the tubes was thus

$$\frac{1}{U_o} = \frac{1}{C_i h_{\text{gni}}} \left(\frac{D_o}{D_i} \right) + R_w + \frac{1}{C_o q_o^n} \quad (2)$$

The conditions for which modified Wilson tests were conducted are summarized in Table 2. These conditions were chosen to cover the conditions that were tested in the subsequent pool boiling and falling film evaporation tests. As the coefficients found with the Wilson plot method are only valid for the conditions tested, it was found necessary to mimic the conditions as exactly as possible. Table 3 shows the values of C_i , C_o and n deduced by this study. The uncertainties quoted are a function of the experimental uncertainties carried through the Wilson plot method using the method of York et al. (2004).

To eliminate all traces of non-condensable gases which might have been introduced into the facility (i.e. during tube or refrigerant changes), a vacuum pump (not shown in Fig. 1) was connected to the system and was run until the two low-pressure reference pressure transducers showed no more than 100 Pa (absolute). Once the vacuum pump was stopped, the system pressure was monitored to make sure that no leaks were present. Only once these two steps were been

Table 2 – Experimental conditions tested for Wilson plot, pool boiling and falling film evaporation experiments.

	Wilson plot	Pool boiling	Falling film
T_{sat}	5–15 °C	5–15 °C	5 °C
$\dot{q}_o$	20–65 kW m ^{−2}	20–80 kW m ^{−2}	20–60 kW m ^{−2}
Re_{wat}	7.5–17.5 · 10 ³	~ 10000	~ 10000
Γ	—	—	0.30–0.02 kg/ms
Refrigerant	R-134a	R-134a/R-236fa	R-134a/R-236fa

Table 3 – Wilson plot coefficients for the Turbo-B5 and the Gewa-B5.

	Turbo-B5	Gewa-B5
C_i	6.6557 ± 0.9117	4.37 ± 0.5398
C_o	10600	$5.72 \cdot 10^6$
n	−0.1129	−0.479

accomplished was the system refilled with refrigerant to proceed with testing. Any remaining traces of non-condensable gases in the system will migrate to the overhead condenser where they remain. The measured saturation temperature using thermocouples and that obtained from the pressure sensors and REFPROP (National Institute of Standards and Technology, 2002) differed by 0.1 K, a value within the uncertainty of both measurements.

For experiments involving overfeed, the film flow rate of the liquid arriving on the first tube was evaluated from the measured mass flow rate and the tube length, assuming that the refrigerant is at saturation conditions. The mass flow of refrigerant evaporating on the first tube is calculated by an energy balance on differential elements and subtracted from the film flow rate arriving on the first tube to obtain the film flow rate at the top of the second tube and so on. This means an ideal one-dimensional downward flow is assumed on the tube rows and assumes that all the liquid flows from one tube to the next without leaving the tube row. The two-pass water design gives a nearly uniform axial film distribution along the tube array after each pair of tubes. The saturation temperature, as well as the transport and thermodynamic properties, are calculated according to REFPROP (National Institute of Standards and Technology, 2002) from the mean of the pressures measured by pressure transducers above and below the tube array.

Tests were conducted by gradually decreasing the liquid film flow rate on the top tube at a fixed heat flux. The data were logged only if steady state conditions were attained. Thermocouples with experimental uncertainty of ± 0.1 K, pressure transducers with uncertainty of ± 0.1 % of full-scale (1 MPa) and Coriolis flow meters (± 0.1 %). An error analysis was performed and the mean uncertainty in the local heat transfer coefficients in pool boiling and falling film are tabulated in Tables 4 and 5. A more detailed description of the test facility, data reduction methods and measurements accuracies can be found in Christians (2010) and Christians et al. (2010a, b).

Table 4 – Experimental uncertainties of measured heat fluxes and pool boiling heat transfer coefficients at all saturation temperatures tested.

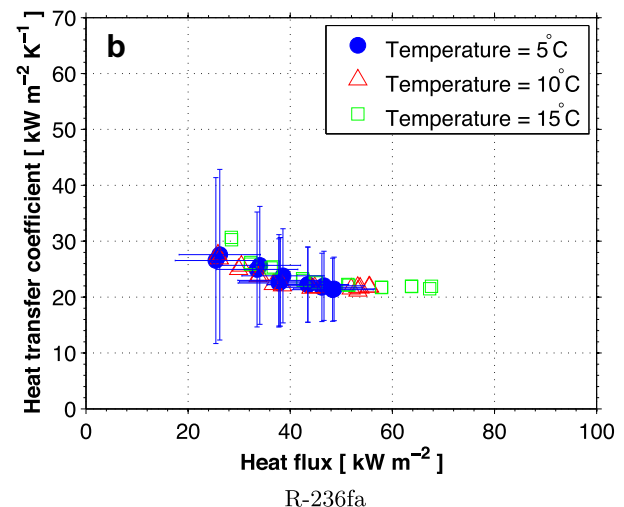
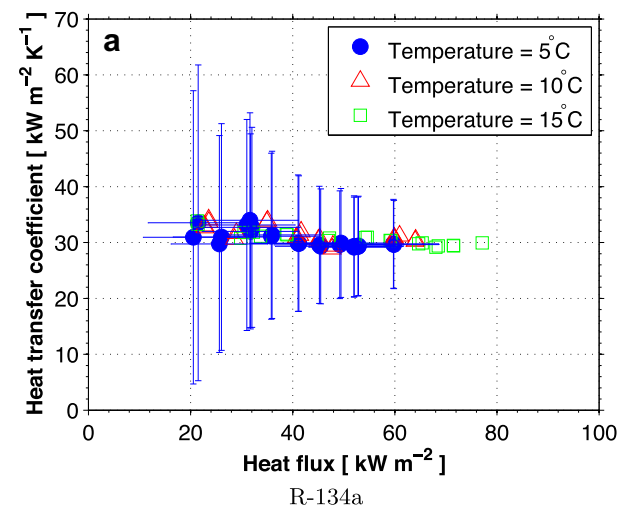
		R-134a			R-236fa		
		$\delta_{\min}$	$\delta_{\max}$	δ_{avg}	$\delta_{\min}$	$\delta_{\max}$	δ_{avg}
Turbo-B5	q	14.49%	47.3%	21.90%	16.08%	31.92%	20.21%
	h_{pb}	26.50%	84.11%	40.73%	18.20%	55.95%	33.27%
Gewa-B5	q	7.75%	25.16%	13.69%	12.33%	34.64%	19.52%
	h_{pb}	18.12%	85.72%	40.88%	19.27%	85.64%	37.98%

Table 5 – Experimental uncertainties of falling film heat transfer coefficients during falling film evaporation.

	q	R-134a		R-236fa	
		$\delta_{\max}$	δ_{avg}	$\delta_{\max}$	δ_{avg}
Turbo-B5	20 kW m ^{−2}	44.97%	31.01%	43.51%	32.72%
	40 kW m ^{−2}	26.01%	19.53%	25.95%	20.27%
	60 kW m ^{−2}	21.91%	15.73%	23.73%	18.77%
Gewa-B5	20 kW m ^{−2}	—	—	—	—
	40 kW m ^{−2}	35.41%	25.53%	35.95%	24.27%
	60 kW m ^{−2}	25.91%	18.73%	23.73%	21.77%

4. Experimental results for pool boiling

Tests were performed using the Wolverine Turbo-B5 and the Wieland Gewa-B5 with R-134a and R-236fa for the experimental conditions summarized in Table 2. Before installation into the test section, the tubes were thoroughly cleaned. The results from the single-array pool boiling tests with tube Turbo-B5 and the Gewa-B5 are detailed in this section. Figs. 2(a), (b) and 3(a), (b) show the results gathered in the

**Fig. 2 – Pool boiling measurements on tube Turbo-B5.**

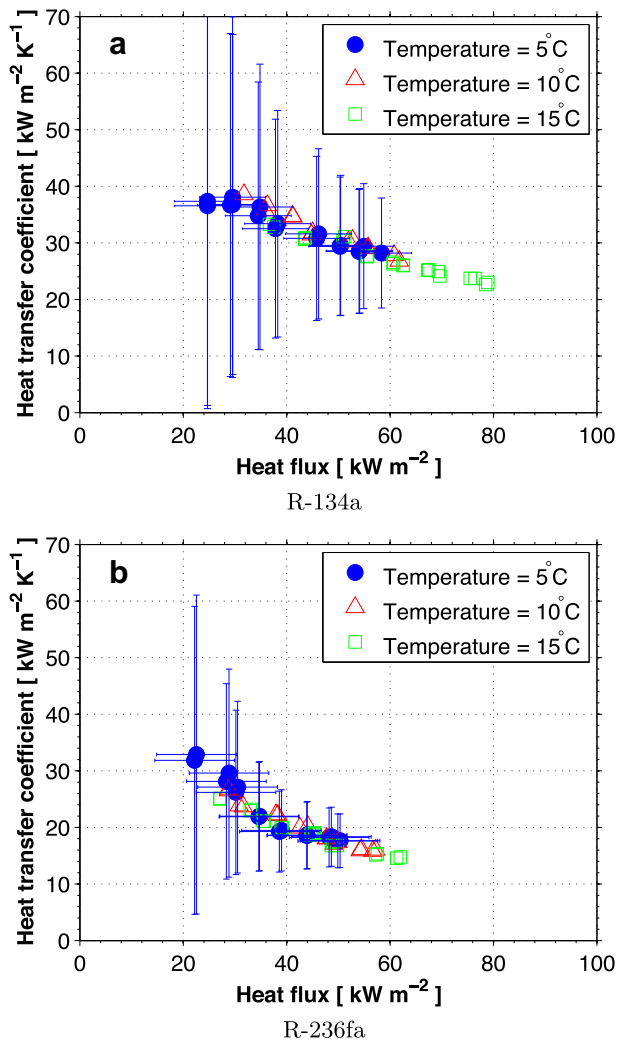


Fig. 3 – Pool boiling measurements on tube Gewa-B5.

experimental conditions proposed using R-134a and R-236fa and tubes Turbo-B5 and Gewa-B5, respectively. Due to industrial interest in low-heat flux data, it was attempted to gather data at heat fluxes lower than 20 kW m⁻². However, due to the decrease in the water temperature difference, the experimental uncertainties became such that it was impossible to gain meaningful values. This can be seen in the error bars plotted in the subsequent figures that increase with decreasing heat flux. The uncertainty increase follows the trend shown in Rose (2004), as expected.

The experimental uncertainties in the heat flux and the measured pool boiling heat transfer coefficients are tabulated in Table 4. The uncertainties in both the heat flux and the measured pool boiling heat transfer coefficient are lowest at the higher heat fluxes tested, due to the increased temperature gradient in the water, as well as the increased difference in temperature between the water side and the saturated refrigerant. The uncertainty in the heat transfer coefficients becomes exponentially worse as the heat flux is decreased. The experimental uncertainties with respect to the refrigerants are essentially the same. Experimentally, it was not possible to reach the lowest heat fluxes using the

Wieland tube, which is why the highest uncertainties shown for this tube are lower than the corresponding Wolverine values.

A comparison with other published results could not be made as no publications with the present new enhanced tube geometries are available in the literature. It should also be noted that these tubes were developed for applications using R-134a and not for R-236fa. Using R-134a, the Turbo-B5 showed almost no dependence on either heat flux nor with different saturation temperatures. The Turbo-EDE2 tested by Habert (2009) (previous generation of the same tube), showed an increased dependence on the saturation temperature; this was not evident in the results of the Turbo-B5. Although at the lower heat fluxes the Turbo-EDE2 tube showed better performance, the high uncertainty in both sets of measurements (Turbo-EDE2 and Turbo-B5) makes it difficult to ascertain the relative performance. However, as the heat flux is increased, the Turbo-B5 can be seen to have increased performance. The Gewa-B5 showed a more marked decrease in heat transfer performance with increasing heat flux, and little dependence on the saturation temperature using R-134a. The Gewa-B4 tested by Habert (2009), showed that when using R-134a, there was little dependence on the saturation temperature as well. In the case of the Gewa-B4, it was found that the heat transfer performance of the tube varied little with increasing heat flux; that is, there was only a relatively small decrease in performance with increasing heat flux (from 48 kW m⁻² K⁻¹–42 kW m⁻² K⁻¹). In the tests performed using the Gewa-B5, the decrease in performance with increasing heat flux is more marked: at a heat flux of 20 kW m⁻², the measured heat transfer coefficient was ~30 kW m⁻² K⁻¹, while at the highest heat flux (80 kW m⁻²), it was ~16 kW m⁻² K⁻¹.

Using R-236fa, the Turbo-B5 and the Gewa-B5 showed lower heat transfer performance, as there is a general decrease in measured heat transfer coefficients when compared to the R-134a results, regardless of the imposed heat flux. There is also a noticeable heat flux dependence, with better performance at the lower heat fluxes tested. This decrease could be due to progressive partial dryout of the enhancement channels. As was the case with R-134a, there does not seem to be any influence of saturation temperature. Comparing the results to those found in the previous experimental campaign using the Gewa-B4, it can be seen that there is a large difference between the two sets of results. In this campaign, the Gewa-B5/R-236fa combination has comparatively worse performance than the Gewa-B5/R-134a combination, including a gentler drop-off to a lower plateau at higher heat fluxes. As was the case with the R-236fa Turbo-EDE2 pool boiling data, the extremely high heat transfer coefficients in the earlier tests seem to be a result of liquid slugs in the pressure transducer capillary tubes leading to the test section; these affected the measured saturation pressure, which led to an incorrect estimation of the saturation temperature and thus the overall heat transfer resistance U_o . This, in turn, resulted in backed out values of the refrigerant-side heat transfer coefficient that were too high.

Smooth-tube pool boiling heat transfer has characteristically shown heat flux exponents ranging from 0.67 to 0.70, as

shown by Stephan (1992), as well as Stephan and Abdelsalam (1980) and Cooper (1984). More recently, this was ratified in this experimental facility by Habert (2009). However, enhanced tubes previously tested in the laboratory have shown a negative exponent for the heat flux, as presented by Robinson and Thome (2004), Roques (2004) and Habert (2009). In this study, when using R-134a, the Turbo-B5 has shown little dependence on the heat flux, with a heat flux exponent that is close to 0 (yet still slightly negative). To validate these results, the same pool boiling tests were performed in the LTCM's bundle boiling facility. The results of this comparison to the data from the bundle boiling facility are shown in Figs. 4 and 5. It can be seen that the data from both facilities are comfortably within each other's uncertainties. The most discrepancy is found at low heat fluxes, where both the Turbo-B5 and the Gewa-B5 show the highest uncertainty in the measurements. It should be mentioned that the actual length of the bundle rig (1000 mm) is roughly twice the length of the falling film active length (500 mm); this makes the experimental uncertainties lower in the bundle boiling facility.

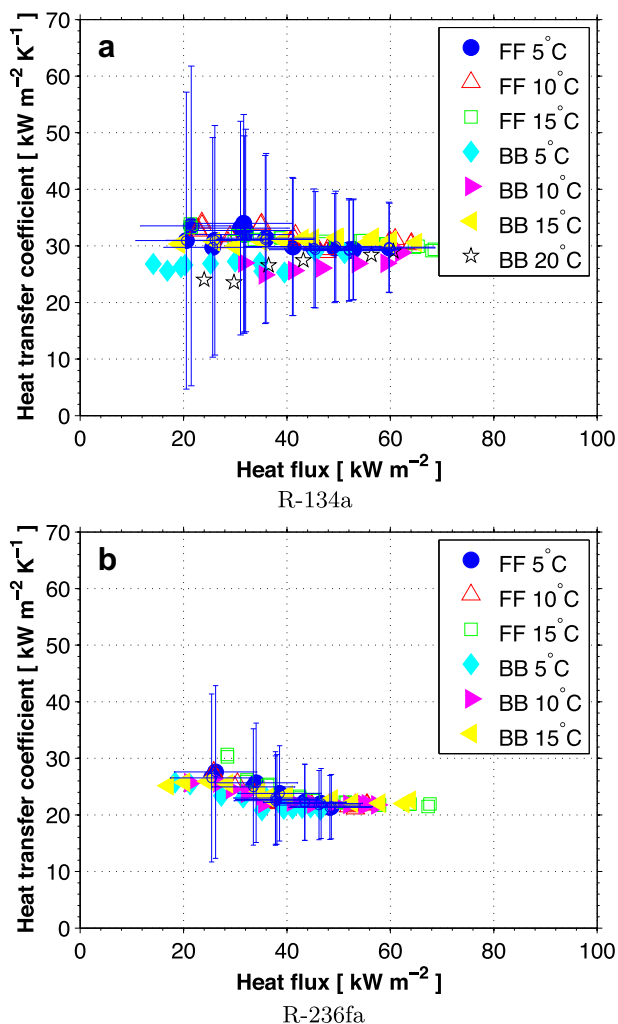


Fig. 4 – Comparison of pool boiling measurements on tube Turbo-B5 performed on the LTCM's falling film and bundle boiling facilities.

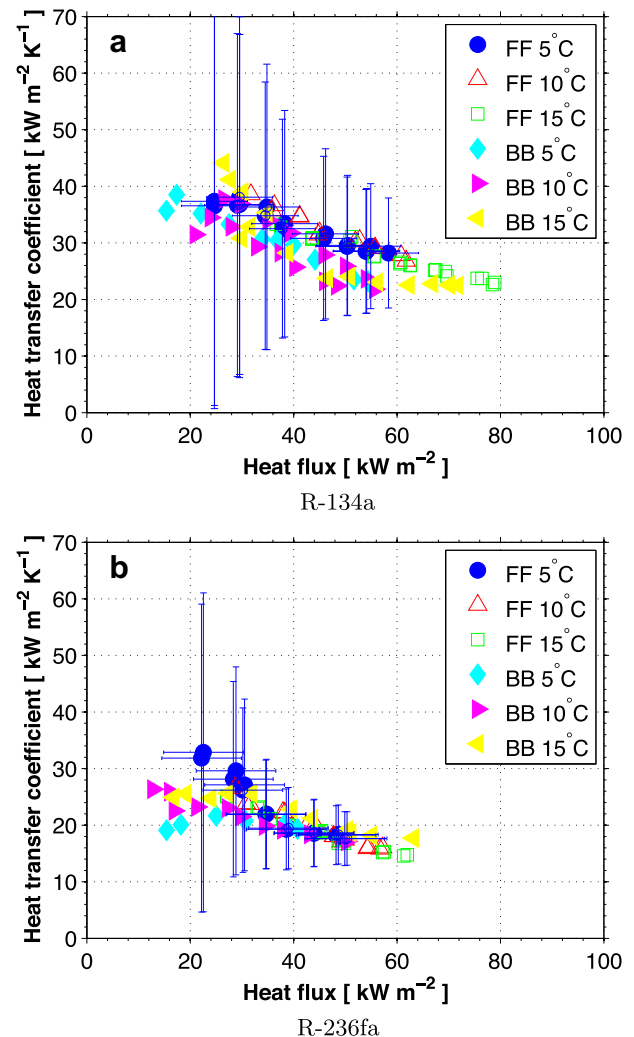


Fig. 5 – Comparison of pool boiling measurements on tube Gewa-B5 performed on the LTCM's falling film and bundle boiling facilities.

5. Experimental results for falling film in single array and in bundle configurations

The experimental conditions are shown in Table 2. For the single-array tests, a single vertical row of horizontal tubes was tested. In bundle conditions, only the center array of ten tubes was instrumented. Furthermore, as in Christians et al. (2010b), the side-array condensate overfeed and side-array heat transfer rates could be modified, thus testing was performed to investigate the effect of the side-array conditions might have had on the measured heat transfer coefficients. As shown in Christians (2010), the different conditions tested did not have any noticeable effect on the measured heat transfer results. Furthermore, it was found that the single-array and the bundle results were the same. Thus, the data from the single-array and bundle tests are presented together for tubes Turbo-B5 and Gewa-B5 in Figs. 6 and 7, respectively.

The data gathered on all ten tubes are only separated by heat flux. Due to the high experimental uncertainties, only data with uncertainties smaller than 45% are shown. This

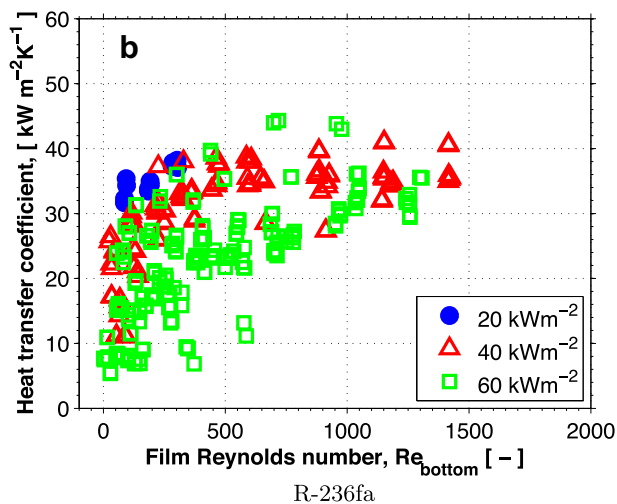
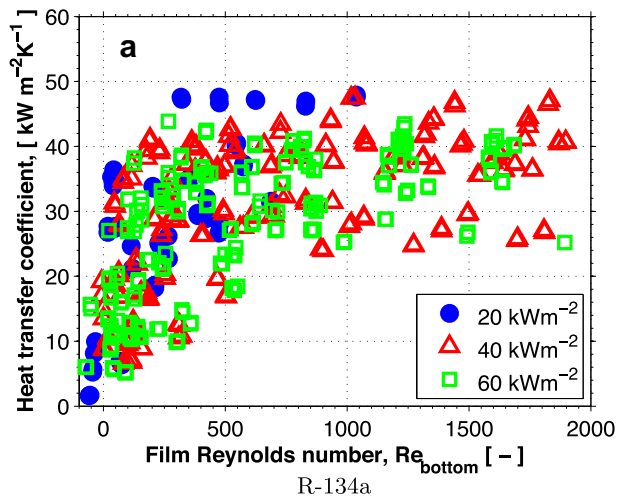


Fig. 6 – Falling film evaporation on a Turbo-B5 tube bundle and single array at $T_{\text{sat}} = 5^\circ\text{C}$.

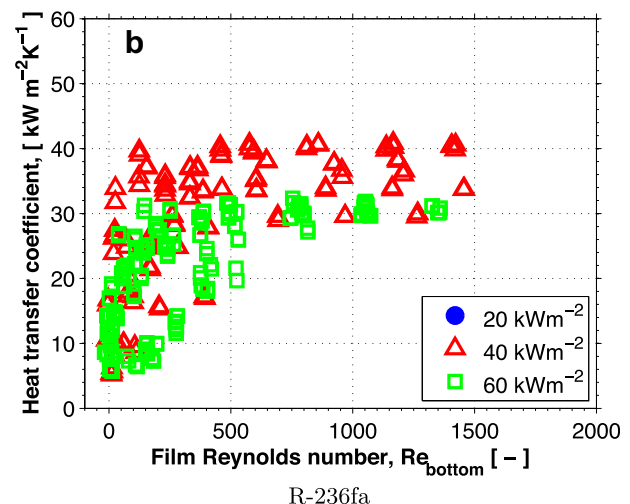
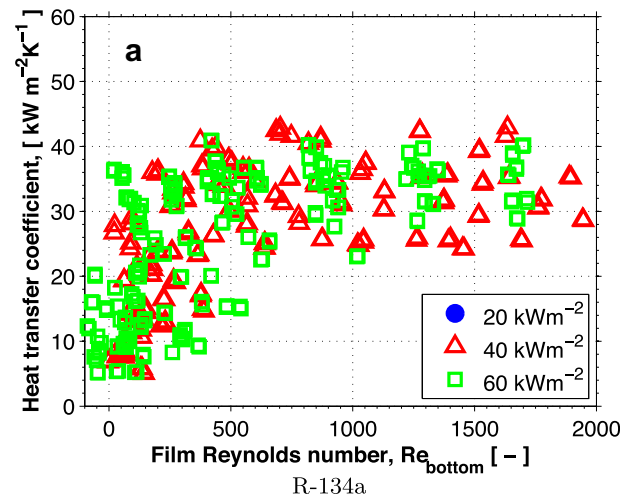


Fig. 7 – Falling film evaporation on a Gewa-B5 tube bundle and single array at $T_{\text{sat}} = 5^\circ\text{C}$.

limit was the result of a parametric study; any higher and the scatter makes trends difficult to discern. The uncertainties during testing are shown in Table 5. The method used to calculate the uncertainties is based on the method of Kline and McClintock (1953), and takes into account measurement, geometric and thermo-physical properties' uncertainties. The method is detailed in Christians (2010). This filtering is why so few 20 kW m^{-2} data are shown; uncertainty values as high as 150% were calculated. As occurred in the pool boiling experiments, the uncertainty in the heat transfer measurements grows exponentially as the heat flux is decreased not only because the temperature gradient in the water is smaller (than at higher heat fluxes), but also due to the fact that the temperature difference between the refrigerant and the water decreases as well. At the moment, for these enhanced tubes and heat fluxes of less than 20 kW m^{-2} , the experimental superheat is on the order of 0.5°C . Keeping in mind that the uncertainty of the thermocouples used to measure the temperatures is $\sim 0.1 \text{ K}$, low heat flux measurements using thermocouples are rapidly reaching their practical limits of usability. The Gewa-B5 required a smaller superheat (than the

Turbo-B5) to initiate boiling, which is why there are no 20 kW m^{-2} data shown in Fig. 7.

Compared to the pool boiling results of Figs. 4 and 5, completely different trends are achieved. First, the progressive heat transfer decrease with increasing heat flux found in all pool boiling experimental data (even the Turbo-B5/R-134a combination had a slight negative trend) is not found in the falling film evaporation data. As described by Roques and Thome (2007a), a heat transfer plateau is reached when the overfeed is large enough such that partial dryout does not occur. At Reynolds numbers lower than this threshold, there is a quasi-monotonic decrease in heat transfer performance. In these results, there does not seem to be a very strong heat flux dependence, as the scatter in the plateau for all heat fluxes tested is large. This results in trends that are difficult to study. From the comparison of the two data sets, it can be seen that the falling film conditions enhanced the heat transfer process, especially in the case of the Turbo-B5. This is shown in Figs. 8 and 9. In these figures, it can be seen that the heat flux has a larger influence on the falling film multiplier, with higher values of K_{ff} for higher heat fluxes. This is of course due

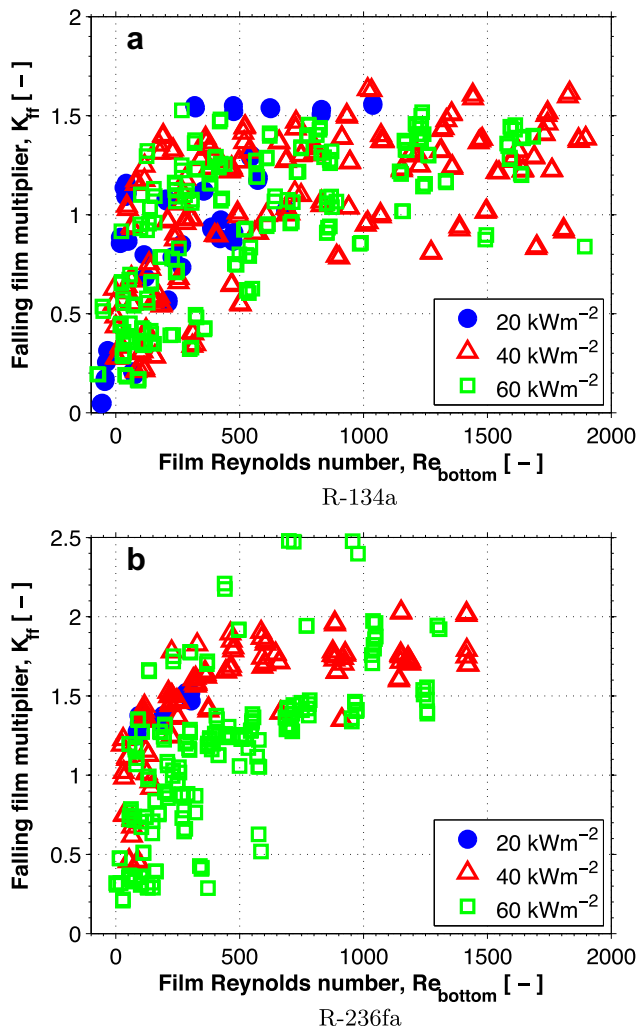


Fig. 8 – Falling film multiplier for the Turbo-B5 tube bundle and single array at $T_{sat} = 5\text{ °C}$.

to the decrease in performance with increasing heat flux seen in pool boiling, a phenomenon not seen in falling film. Of interest is that while in pool boiling there was a marked decrease in heat transfer performance when using R-236fa, there is a much smaller ‘penalty’ (and at times, no penalty at all) when in falling film evaporation mode. A hypothesis for this is discussed based on the visualizations in Christians and Thome (2010). This decrease is also the reason why R-236fa falling film multipliers are comparatively larger than R-134a multipliers on the same tube. From these results it could be hypothesized that, in a low pressure environment, the falling film process is not only advantageous from a charge reduction point of view. The decrease in performance typically seen when designing for low pressure refrigerants can be seen to be seriously mitigated; in fact, the results show that the use of falling film allowed the heat transfer performance in R-236fa to be very close, if not the same, to the performance of R-134a of these tubes. An added advantage of using falling film in low pressure refrigerants is that the submergence effects are also decreased (due to not having a large liquid level in the evaporator).

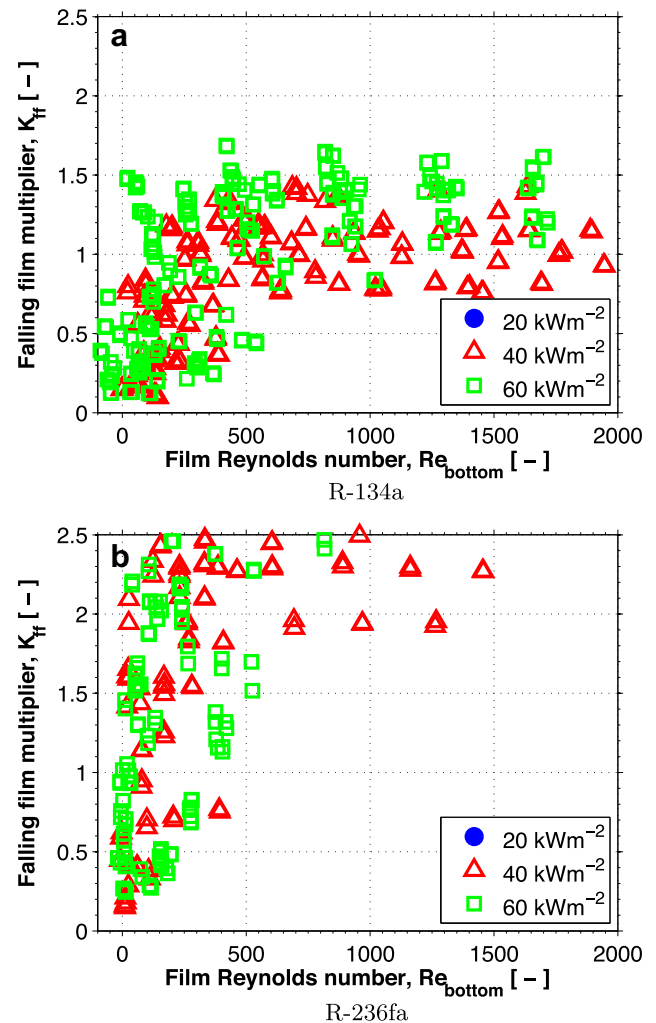


Fig. 9 – Falling film multiplier for the Gewa-B5 tube bundle and single array at $T_{sat} = 5\text{ °C}$.

As the reader may recall, the refrigerant film Reynolds numbers are calculated assuming one-dimensional flow from one tube to the next. However, during experimentation, some tubes showed splash-out and slinging. This loss of refrigerant to the side-arrays (when in bundle mode) is not quantifiable, and thus errors are introduced into the presented Reynolds numbers when some liquid leaves the film. This is the main culprit for some of the scatter evidenced in the data which happen to have lower heat transfer performance than expected at specific Reynolds numbers. This is evident in the R-236fa data shown in Figs. 6 and 7, in which data at relatively large Reynolds numbers show low heat transfer performance. Since the Reynolds numbers presented are calculated and are dependent on the basic assumption that there is no splash-out, when this assumption is no longer valid an error is introduced into the calculated Reynolds numbers such that data are presented at higher film Reynolds numbers in the cases of ‘splash out’ than is actually the case.

In the tests performed by Habert (2009), some of the data for tubes Turbo-EDE2 and Gewa-B4 (and within similar uncertainty ranges) showed results on the order (or slightly

smaller) than those measured in this study Christians (2010). The general trends of a plateau, a threshold Reynolds number and decreasing heat transfer performance with increasing dryout were also found in that study. There was also little influence of the heat flux on the falling film heat transfer performance. Unlike his study, however, no other types of bundle effects were found in the results presented here. For enhanced boiling tubes, Habert (2009) found large mal-distribution down the tested bundles (identical in layout and geometry as the presented results, the only difference being the tubes tested), with some tubes ‘starved’ of overfeed.

6. Onset-of-dryout

The onset-of-dryout was first experimentally characterized at the LTCM laboratory by Roques (2004), in a separate set of experiments in which the film flow rate was decreased in small steps from a high value at a fixed heat flux. The transition from fully wet to dryout was found to occur at a film flow rate just at the first dryout patch formation. A simple linear correlation was made. Later, the criterion based on K_{ff} defined by Ribatski and Thome (2007) was used to more objectively study the film breakdown (Ribatski and Thome, 2007; Habert, 2009), as the method first utilized by Roques (2004) was dependent on a subjective visual analysis of the flow over a tube. The criterion is defined as

$$-0.10 < \frac{K_{ff,j} - \frac{1}{n} \sum_{j=1}^n K_{ff,j}}{\frac{1}{n} \sum_{j=1}^n K_{ff,j}} < -0.05 \quad (3)$$

The lower boundary of -0.10 was chosen such that the filtering program did not add data that were clearly in partially dry conditions into the database utilized to generate a prediction method for the onset-of-dryout. The data were also filtered with a blanket filter that disregarded any data with outside heat transfer coefficient uncertainties higher than 45%, as described above. The simple dryout criteria method of Roques (2004), in which the threshold Reynolds number (i.e. the onset-of-dryout Reynolds number) was assumed to be two times the dryout Reynolds number, compares well to the method proposed by Ribatski and Thome (2007) and implemented here, giving results within 15% of each other for the current experimental database.

The data from the two tubes studied in the present experimental campaign, filtered according to the modified Ribatski criteria, are presented in Fig. 10(a) and (b). Refer to Christians (2010) for the complete onset-of-dryout database based on all the enhanced boiling tube/refrigerant combinations tested at the LTCM laboratory.

The definition of ‘complete evaporation line’ is the same as that of Roques (2004), namely, it shows the required refrigerant for all of the impinging liquid to be evaporated at a specific heat flux. The results show a larger amount of scatter than those of Roques (2004) because, while the criterion defined by Ribatski and Thome (2007) works well, the

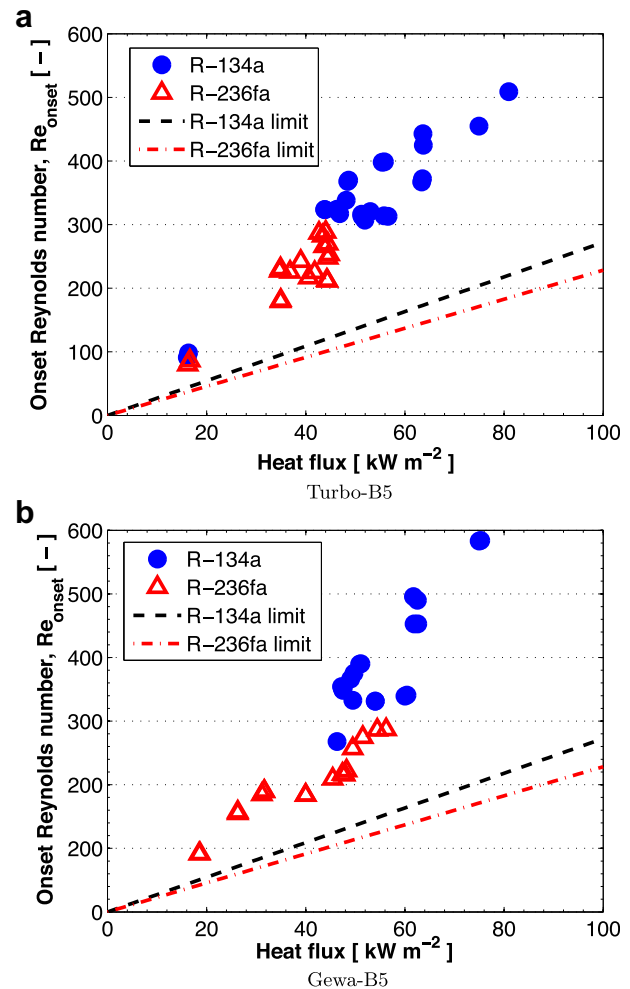


Fig. 10 – Onset-of-dryout Reynolds numbers for the enhanced tubes of the current study.

errors and uncertainties in the calculation of K_{ff} are propagated into this calculation as well.

The experimental data of the Gewa-B5 are close to those measured using the Gewa-B4. The same can be said of the results found using the Turbo-B5, when comparing it to the previous generation Turbo-EDE2. For all tubes, the onset-of-dryout Reynolds number increases with heat flux. That is, a higher film flow rate is required to avoid onset-of-dryout as the heat flux rises. While the Turbo-Bii HP and the Gewa-B2 maintained a reasonably constant distance from the complete evaporation line (that is, ran parallel to it), the Turbo-EDE2, Turbo-B5, Gewa-B4 and the Gewa-B5 all show non-parallel behavior. The vertical distance between the complete evaporation line and the onset-of-dryout Reynolds number corresponds to the Reynolds number at the bottom of the tube. Thus, the maximum of the partially dry heat transfer coefficient data depend slightly on the heat flux. This is most evident when plotting the heat transfer coefficient versus Reynolds number: this difference is the cause of the spread at the threshold of wet/partially dry data, evidenced in Figure 6.4 of Roques (2004) and Figure 6.4 of Christians (2010), among others.

7. Conclusion

Experimental data collected for tubes Turbo-B5 and Gewa-B5 using refrigerants R-134a and R-236fa in both pool boiling and falling film tests were presented and discussed. The trends in the pool boiling results were compared to those found in previous studies for earlier-generations of these tubes. In pool boiling, the tubes performed better than the tubes tested by Roques (2004) and Habert (2009), and showed less dependence on the applied heat flux. The falling film heat transfer data were presented for both tubes and refrigerants utilized. No additional bundle effects could be found when testing in bundle configuration compared to single array tests. A falling film multiplier, relating the increase in heat transfer performance when using falling films versus pool boiling was calculated for the falling film measurements and presented. In terms of refrigerants, it was seen that R-134a outperformed R-236fa, as was expected, since these tubes are designed for use with R-134a. However, the degradation was smaller when testing in falling film than in pool boiling. The onset-of-dryout threshold Reynolds numbers were presented for the two tubes and refrigerants tested. These experimental data show that low heat flux measurements using thermocouples are nearing the limits of their applicability, since the majority of the data are too uncertain to be used. This forces studies of next-generation tubes to both start using high accuracy RTDs (a challenge for the instrumentation used at the LTCM) and to start taking advantage of differential RTD temperature measurements, which retain the uncertainty of a single RTD (± 0.03 K). The data presented here are utilized in part 2 of Christians and Thome (2010) to generate new prediction methods.

Acknowledgments

This project has been supported financially by the LTCM Falling Film Research Club members: Carrier Corporation, the Trane Company, Wieland-Werke AG and Wolverine Tube Inc., which is gratefully acknowledged. Special acknowledgment is made to Wieland-Werke AG and Wolverine Tube Inc. for providing the tube samples for the tests.

REFERENCES

- Agostini, F., 2008. Boiling on a Tube Bundle: Heat Transfer, Pressure Drop and Flow Patterns. Ph.D. thesis. Ecole Polytechnique Fédérale de Lausanne, Laboratory of Heat and Mass Transfer, Lausanne, Switzerland.
- Bergles, A.E., 1988. Heat Transfer Augmentation. In: Two-Phase Flow Heat Exchangers, pp. 343–373.
- Cerza, M., Sernas, V., 1985. A bubble growth model for nucleate boiling in thin, falling, superheated, laminar, water films. *Int. J. Heat Mass Transfer* 28, 1307–1316.
- Chien, L.H., Webb, R.L., 1998a. A parametric study of nucleate boiling on structured surfaces, part I: effect of tunnel dimensions. *J. Heat Transfer* 120, 1042–1048.
- Chien, L.H., Webb, R.L., 1998b. A parametric study of nucleate boiling on structured surfaces, part II: effect of pore diameter and pore pitch. *J. Heat Transfer* 120, 1049–1054.
- Chien, L.H., Webb, R.L., 1998c. Visualization of pool boiling on enhanced surfaces. *Exp. Therm. Fluid Sci.* 16, 332–341.
- Christians, M., 2010. Heat transfer and visualization of evaporation on a tube bundle. Ph.D. thesis. Ecole Polytechnique Fédérale de Lausanne, Laboratory of Heat and Mass Transfer, Lausanne, Switzerland.
- Christians, M., Thome, J.R., 2010. Falling film evaporation on enhanced tubes, part 2: Prediction methods and visualization. *Int. J. Refrigeration*, submitted for review.
- Christians, M., Habert, M., Thome, J.R., 2010a. Film condensation of R-134a and R-236fa, part 1: experimental results and predictive correlation for single-row condensation on enhanced tubes. *Heat Transfer Eng.* 31, 799–808.
- Christians, M., Habert, M., Thome, J.R., 2010b. Film condensation of R-134a and R-236fa, part 2: experimental results and predictive correlation for bundle condensation on enhanced tubes. *Heat Transfer Eng.* 31, 809–820.
- Chyu, M.C., Bergles, A.E., 1985. Falling film evaporation on a horizontal tube. In: *Multiphase Flow and Heat Transfer*, vol. 43. ASME HTD, pp. 39–48.
- Cooper, M.G., 1984. Heat flow rates in saturated nucleate pool boiling - a wide-ranging examination using reduced properties. *Adv. Heat Transfer* 16, 157–239.
- Danger, E., Chandra, U., Gorenflo, D., Beutler, A., 2001. Basic modifications of the heated surface for enhancement of pool boiling heat transfer. In: *Thermophysical Properties and Transfer Process of New Refrigerants*, pp. 380–387.
- Gnielinski, V., 1975. New equations for heat and mass transfer in turbulent flow through pipes and ducts. *Forschung Ing. Ingenieurwesen* 41, 1975.
- Gorgy, E.I., 2008. Pool boiling of R-134a and R-123 on smooth and enhanced tubes. Master's thesis. Kansas State University, Department of Mechanical and Nuclear Engineering.
- Habert, M., 2009. Falling Film Evaporation on a Tube Bundle with Plain and Enhanced Tubes. Ph.D. thesis. Ecole Polytechnique Fédérale de Lausanne, Laboratory of Heat and Mass Transfer, Lausanne, Switzerland.
- Habert, M., Thome, J.R., 2010. Falling film evaporation on tube bundle with plain and enhanced tubes-part I: experimental results. *Exp. Heat Tran.* 23 (4), 259–280.
- Hahne, E., Barthau, G., 2006. Heat transfer and nucleation in pool-boiling. *Int. J. Therm. Sci.* 45, 209–216.
- Hsieh, S.S., Hsu, P.T., 1994. Nucleate boiling characteristics of R-114, distilled water and R-134a on plain and rib-roughened tube geometries. *Int. J. Heat Mass Tran.* 37, 1423–1432.
- Kline, S.J., McClintock, F.A., 1953. Describing uncertainties in single-sample experiments. *Mech. Eng.* 75, 3–8.
- Kramer, W., Auracher, H., Marvillet, C., 1997. Cross-flow and pool boiling of ethanol-water mixtures on the outside of horizontal enhanced tubes. In: *3rd Conference of the International Engineering Foundation*, pp. 419–426.
- McNeil, D.A., Burnside, B.M., Miller, K.M., Tarrad, A.H., 2002. A comparison between high flux and plain tubes for boiling pentane in a horizontal kettle reboiler. *Appl. Therm. Eng.* 2, 803–814.
- Memory, S.B., Sugiyama, D.C., Marto, P.J., 1995. Nucleate pool boiling of R-114 and R-114-oil mixtures from smooth and enhanced surfaces—I. single tubes. *Int. J. Heat Mass Transfer* 38, 1347–1361.
- Moeykens, S.A., Pate, M.B., 1994. Spray evaporation heat transfer of R-134a on plain tubes. *ASHRAE Trans.* 100, 173–184.
- Moeykens, S.A., Huebsch, W.W., Pate, M.B., 1995. Heat transfer of R-134a in single tube spray evaporation including lubricant effect and enhanced surface results. *ASHRAE Trans.* 101, 111–123.
- National Institute of Standards, Technology, 2002. NIST Thermodynamic Properties of Refrigerants and Refrigerant Mixtures Database. Ver 8.0 (Gaithersburg, MD).

- Poniewski, M., Thome, J.R., 2008. Nucleate Boiling on Micro-Structured Surfaces. www.htri-net.com/ePubs/epubs.htm.
- Ribatski, G., Jacobi, A.M., 2005. Falling-film evaporation on horizontal tubes—a critical review. *Int. J. Refrigeration* 28, 635–653.
- Ribatski, G., Thome, J.R., 2006. Nucleate boiling heat transfer of R-134a on enhanced tubes. *Appl. Therm. Eng.* 26, 1018–1031.
- Ribatski, G., Thome, J.R., 2007. Experimental study on the onset of local dryout in an evaporating falling film on horizontal plain tubes. *Exp. Therm. Fluid Sci.* 31, 483–493.
- Robinson, D., Thome, J.R., 2003. Flooded Evaporation Heat Transfer Performance Investigation for Tube Bundles Including the Effects of Oil Using R-410A and R-507A Technical Report. ASHRAE Research Project RP-1089.
- Robinson, D., Thome, J.R., 2004. Local bundle boiling heat transfer coefficients on a Turbo-BII HP enhanced tube bundle. *HVAC&R Res.* 10, 441–457.
- Roques, J.F., 2004. Falling Film Evaporation on a single tube and on a tube bundle. Ph.D. thesis. Ecole Polytechnique Fédérale de Lausanne, Laboratory of Heat and Mass Transfer, Lausanne, Switzerland.
- Roques, J.F., Thome, J.R., 2007a. Falling films on arrays of horizontal tubes with R-134a, part I: boiling heat transfer results for four types of tubes. *Heat Transfer Eng.* 28, 398–414.
- Roques, J.F., Thome, J.R., 2007b. Falling films on arrays of horizontal tubes with R-134a, part II: flow visualization, onset of dryout, and heat transfer predictions. *Heat Transfer Eng.* 28, 415–434.
- Rose, J.W., 2004. Heat-transfer coefficients, wilson plots and accuracy of thermal measurements. *Exp. Therm. Fluid Sci.* 28, 77–86.
- Stephan, K., 1992. *Heat Transfer in Condensation and Boiling*. Springer, Berlin.
- Stephan, K., Abdelsalam, M., 1980. Heat transfer correlation for natural convection boiling. *Int. J. Heat Mass Transfer* 23, 73–87.
- Tatara, R.A., Payvar, P., 2000. Pool boiling of pure R-134a from a single turbo-BII-HP tube. *Int. J. Heat Mass Transfer* 43, 2233–2236.
- Thome, J.R., 1990. *Enhanced Boiling Heat Transfer*. Hemisphere Publishing Corporation.
- Thome, J.R., 2002. On Recent Advances in Modeling of Two-Phase Flow and Heat Transfer Keynote address: 1st International Conference on Heat Transfer, Fluid Mechanics, and Thermodynamics 8 - 10 April, 13 pages.
- van Rooyen, E., Christians, M., Thome, J.R., 2010. Modified Wilson plots for enhanced heat transfer experiments: Current status and future perspectives. *Heat Transfer Engineering*, submitted for publication.
- Webb, R.L., Pais, C., 1992. Nucleate pool boiling data for five refrigerants on plain, integral-fin and enhanced tube geometries. *Int. J. Heat Mass Transfer* 35, 1893–1904.
- York, D., Evensen, N.M., Martinez, M.L., Delgado, J.D.B., 2004. Unified equations for the slope, intercept, and standard errors of the best straight line. *Am. J. Phys.* 72, 367–375.
- Zeng, X., Chyu, M.C., Ayub, Z.H., 1996. Evaporation heat transfer performance of nozzle-sprayed ammonia on a horizontal tube. *ASHRAE Trans.* 101, 136–149.